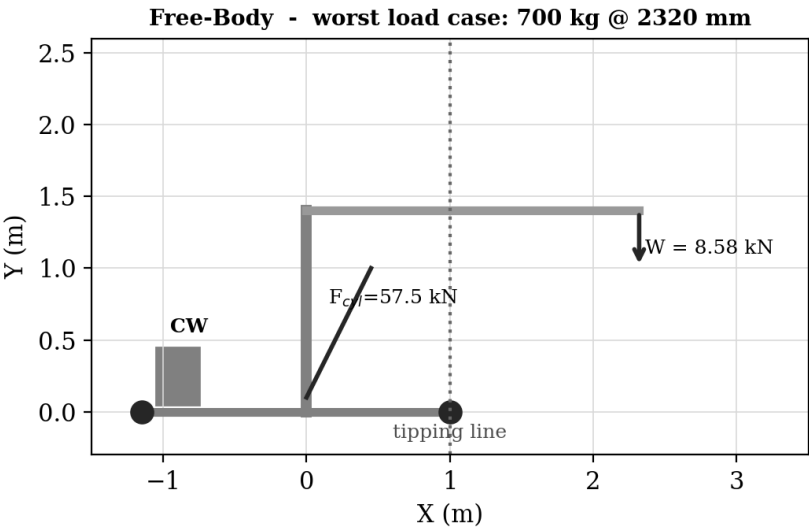


PROJECT 40467361

STRUCTURAL CALCULATION AND
LOAD VERIFICATION REPORT

1 Tonne Mobile Floor Jib Crane

Combined verification: wheels-only and deployed-outrigger configurations, on Q235 material



Free body at the governing case, 700 kg at 2320 mm. The tipping line is at the front wheel, 980 mm ahead of the column.

Field	Detail
Project reference	40467361
Configurations assessed	Outriggers stowed (wheels only) and outriggers deployed (# arrangement)
Material	Q235 carbon structural steel (fy 235 MPa)
Rated capacity (as marked)	1000 kg at 1400 mm reach
Document revision	Rev 5 (combined; supersedes Rev 4 on the Q235 material and 2500 mm track)
Date	12 June 2026
Status	Issued for review
Analysis tool	MATLAB R2026a (parametric verification model, project 40467361)
Companion document	Load Re-Rating Report (Rev R2), issued separately

Document Control

This report is the combined independent verification of the 1 tonne mobile jib crane shown on the manufacturer drawing set. It assesses the crane in both of its working stances in a single document: standing on its four wheels with the outrigger legs stowed, and standing on its four deployed legs in the # arrangement. This revision, Rev 5, carries the crane material as Q235 (not S275) and takes the deployed lateral track from the updated general-arrangement drawing at 2500 mm. The two configurations differ only below the slewing bearing, in stability and in the members that carry the load to the ground; everything above the bearing is identical in both stances.

Revision history

Rev	Date	Description	Status
2	09 Jun 2026	Wheels-only detailed report (S275).	Superseded
A	10 Jun 2026	Deployed-outrigger study (S275, 1200 mm front line).	Superseded
3	10 Jun 2026	Combined verification, S275, 980 mm front line.	Superseded
4	11 Jun 2026	Combined, B&W figures and verdicts corrected.	Superseded
5	12 Jun 2026	Material set to Q235 (fy 235 MPa, was S275); deployed track 2500 mm from the updated drawing. Stress-based factors revised down; outrigger arm and boom outer now fail; sideways stability revised. Proof-load references removed.	Issued for review

Basis and limitations

- Material. The crane material is Q235 (GB/T 700), fy 235 MPa, confirmed by the client. All sections are 6 mm wall, so the full 235 MPa applies with no thickness de-rating. The member allowable is $235 / 1.5 = 156.7$ MPa. Q235 is the equivalent of ASTM A36 / EN S235JR.
- Drawings as issued. Section sizes, member lengths, weld throats, bolt grade, the slewing-bolt pattern and the leg dimensions are read directly off the supplied drawings. The deployed track is taken from the updated general-arrangement plan view at 2500 mm foot-to-foot.
- Self-weight. The crane mass of 700 kg is the client figure. The mass split and CG offset are engineering estimates, flagged in Section 26.
- Deployed geometry. The legs are taken square to the rails at 90 degrees. The deployed feet land at the 2500 mm track shown on the drawing; the as-built width should be confirmed.
- Scope of physics. Static lifting only. Wind, seismic, side pull and snatch are outside scope. A 1.25 dynamic factor covers normal controlled hoisting per HC1 duty.
- Not a design. This verifies an existing unit against its rated chart in both stances. It does not replace load testing or a competent-person examination, which are carried out separately with the approval officer.

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1. Executive Summary

The crane was checked against its rated load chart in both working stances, on the Q235 material. Fourteen structural and mechanical checks were carried out, plus stability in each stance, by two independent methods: closed-form hand calculation and an in-script plane-frame finite element model. The central finding is consistent across both stances and is easy to act on.

The crane does not hold the working-stress margins of a rated 1 tonne appliance, in either stance, and on Q235 the shortfall is deeper than the earlier S275 study showed. Of the fourteen structural and mechanical checks, eight fall short and both stability checks also fall short. The shortfalls cluster on one operating point and one load path. The governing case is not the heaviest lift but the 700 kg load held out at 2320 mm, which produces the largest moment on the structure, 19.91 kN·m. At that point the slewing column reaches its yield stress (factor of safety 0.85) and the fillet weld at its base is worse again, at 0.73. These two items are the real headline; the boom outer section and the base outrigger arm, which were marginal or passing on S275, now also fail on the weaker steel.

1.1 Wheels-only configuration

Standing on its wheels, the crane will pick the loads up, but on Q235 the margins are thin: the column reaches yield, its base weld is at 0.73, and the boom outer and outrigger arm now fall below target. It cannot slew safely on its wheels, because the narrow rail track is the whole footprint. The column and its base weld, the boom outer section, the outrigger members, the pivot and cylinder pins, and forward stability all fall short of target.

1.2 Deployed-outrigger configuration

Deploying the four legs in the # arrangement is what lets the crane slew, by widening the lateral track to 2500 mm. But the legs sit at 90 degrees to the rails, so they reach sideways and not forward. Forward tipping is therefore unchanged from the wheels-only case and still fails at mid reach (0.96). Sideways tipping, the case the legs are there for, clears the target only close in; at mid and far reach it falls to 1.17 and 1.42 on the 2500 mm track, and reaches target at about 2784 mm. The deployed leg member itself passes at 1.64. Crucially, nothing above the slewing bearing changes: the column stays at 0.85 and its base weld at 0.73 in both stances.

1.3 Headline results

Governing load case, 700 kg at 2320 mm. Stability values are the worst across the chart in each stance.

Item	Result	FoS	Target	Verdict
Column, combined N+M (both stances)	277 MPa	0.85	1.50	Fail
Column-to-base weld (both stances)	235 MPa	0.73	1.50	Fail
Boom outer section (both stances)	186 MPa	1.26	1.50	Fail
Outrigger arm bending (wheels)	180 MPa	1.30	1.50	Fail
Cylinder rod-end pin (Ø20 as modelled)	92 MPa	1.03	3.00	Fail
Forward stability (both stances)	9.20 kN·m	0.96	1.50	Fail
Sideways stability, slewing (2500 track)	7.35 kN·m	1.17	1.50	Fail
Deployed leg member (RHS 60x100x6)	144 MPa	1.64	1.50	Pass

Item	Result	FoS	Target	Verdict
Column buckling, Pcr	720 kN	67.6	1.50	Pass
Slewing bolts, utilisation 0.35	—	1.69	1.50	Pass

Plastic reserve. The column reaches first-fibre yield near the rated load, but as a compact section it holds reserve beyond first yield before a plastic hinge can form, which the elastic check does not credit; Section 17 sets this out. The factor of 0.85 is a first-yield working-stress result rather than a collapse prediction, but the structure still does not hold the conventional 1.5 working-stress margin for sustained duty.

Way forward. The crane cannot be certified to the full 1 tonne chart on a working-stress basis in either stance. The recommended action is to re-rate the chart to what the Q235 steel safely carries, which needs no fabrication; the separate Load Re-Rating Report (Rev R2) sets a maximum SWL of 560 kg with reduced values at reach. Retaining the full 1 tonne rating requires the larger column section and matching base weld. Deploying the legs is a precondition for slewing, not a substitute for that structural work.

PART I — Common basis (applies to both configurations)

2. Introduction and Scope

2.1 Background

The unit is a mobile floor jib crane built to lift up to 1 tonne. It runs on four wheels and carries a counterweight at the rear. A telescopic boom in three stages luffs on a single hydraulic cylinder, and the whole upper assembly slews on a bolted bearing flange above the base frame. The rated load chart marks three points: 1000 kg at 1400 mm, 700 kg at 2320 mm and 300 kg at 3300 mm. The warning plate instructs the operator to deploy the legs before slewing.

The client commissioned an independent check to confirm whether the structure carries that chart with adequate margin against yielding, buckling, weld and pin failure, and overturning, in both stances, on the Q235 material. The work was scoped as a calculation exercise.

2.2 What this report covers

For the wheels-only stance the verification addresses eight areas, each reported with a factor of safety and a pass or fail against target: boom bending; column stress plus buckling; base-frame wheel reactions and outrigger-arm bending; pin shear and bearing; fillet-weld strength; the slewing-flange bolt check; stability at every chart point; and a combined statement on the 1 tonne safe working load. For the deployed stance it adds forward and sideways stability, the deployed leg-member check, and a statement of which wheels-only results carry over.

2.3 Configurations assessed

Both stances are assessed in full. Where a result is identical in both, it is derived once and noted as common. The column and base-weld findings, in particular, stand in either configuration, because they sit above the slewing bearing and are fixed by the boom and the lifted load.

3. Materials, Sections and Allowable Stresses

3.1 Material and target factors of safety

All structural members are hollow steel sections in grade Q235 (GB/T 700), confirmed by the client. The yield stress is $f_y = 235$ MPa, with an ultimate of 370 to 500 MPa; the minimum 370 MPa is used for weld checks. All sections are 6 mm wall, so the full 235 MPa applies. Plates are A36 or equivalent, bolts are grade 8.8, and fillet welds are 6 mm throat minimum. Members are checked against yield with a target of 1.5; pins and bolts in shear and bearing carry a target of 3.0.

Allowable stresses (Q235, $f_y = 235$ MPa, $f_u = 370$ MPa)

Member bending / direct stress (target FoS 1.5):

$$\sigma_{\text{allow}} = f_y / 1.50 = 235 / 1.50 = 156.7 \text{ MPa}$$

Pin / bolt shear (target FoS 3.0):

$$\tau_{\text{allow,pin}} = 0.6 \times f_y / 3.0 = 0.6 \times 235 / 3.0 = 47.0 \text{ MPa}$$

Fillet weld (on throat, as coded):

$$\tau_{\text{allow,weld}} = 0.7 \times f_u / 1.5 \quad (f_u = 370 \text{ MPa}) = 172.7 \text{ MPa}$$

3.2 Section properties

Section properties for a rectangular hollow section come from the standard expressions, where B and H are the outer width and depth and t the uniform wall thickness. The same expressions applied to every member give the properties used throughout, which match the values printed by the analysis model.

Member	Section (RHS)	A (cm ²)	I _x (cm ⁴)	S _x (cm ³)
Boom outer (root)	100 x 140 x 6	27.4	748.8	107.0
Boom middle	80 x 120 x 6	22.6	438.2	73.0
Boom inner (tip)	60 x 100 x 6	17.8	227.4	45.5
Column	80 x 120 x 6	22.6	438.2	73.0
Base frame / outrigger / leg	60 x 100 x 6	17.8	227.4	45.5

4. Methodology

Two independent methods were run against the same geometry so the results could be cross-checked. The first is closed-form hand calculation following Eurocode 3 and EN 13001. The second is a small two-dimensional frame solver written inside the model. Both read their inputs from one parametric block, so any change to a section size, material grade or weld throat flows through every check at once.

4.1 Track 1: closed-form hand calculation

Each member is reduced to its governing internal action and checked by hand. Bending uses the elastic section modulus, axial load uses the gross area, and the two combine linearly for the column. Pins are checked in double or single shear and in bearing. Welds are checked on the throat area. Stability is a moment balance about the tipping line.

4.2 Track 2: in-script frame finite element check

The crane is modelled as a plane frame of Euler-Bernoulli beam elements, each carrying a 6 by 6 stiffness matrix rotated into the global frame and assembled into the system stiffness. The front wheel is pinned and the rear is a roller. The frame is re-solved for all three chart points because the boom extends to a different length at each. The base and outrigger elements use twice the area and second moment, to lump the two side rails into one equivalent beam.

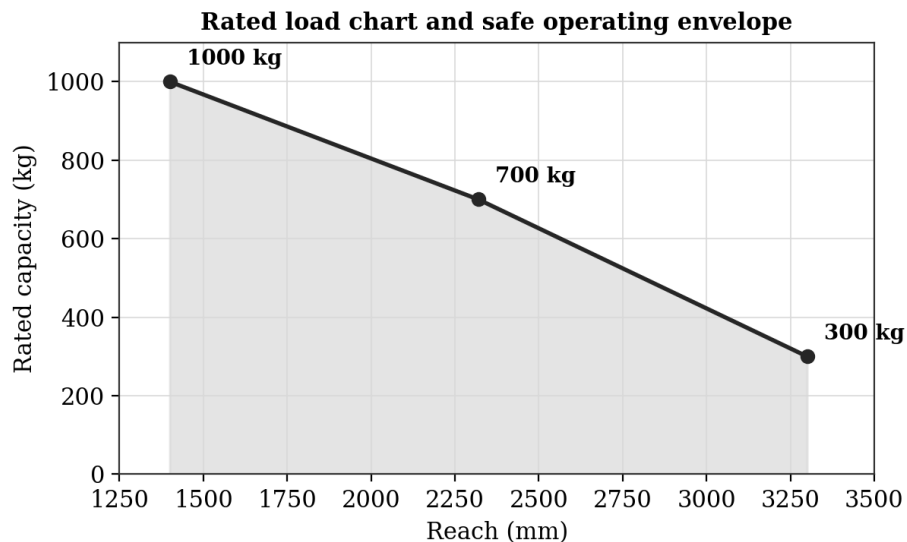
4.3 Cross-checking the two tracks

The two methods are expected to agree on the column-base moment and the peak member stress. Where they differ is informative: the hand calc treats the outrigger as a simple cantilever, while the frame model captures moment transfer through the bolted joint. Section 16 reports the comparison.

5. Load Model and Governing Case

5.1 Rated load chart

The manufacturer chart gives three rated points; capacity falls as reach increases. A dynamic amplification factor of 1.25 is applied to the static load, consistent with HC1 hoist duty under EN 13001.



Rated load chart with the safe operating envelope shaded.

Chart point	Reach (mm)	Rated load (kg)	Static load (kN)	Factored load (kN)
Close	1400	1000	9.81	12.26
Mid	2320	700	6.87	8.58
Far	3300	300	2.94	3.68

5.2 Which case governs

The design moment at the column is the factored load times its reach. The mid-reach case is the most severe, even though it is not the heaviest load.

Governing design moment

Design moment at column, $M = W \times r$

Close 1000 kg: $M = 12.26 \text{ kN} \times 1.400 \text{ m} = 17.17 \text{ kN.m}$

Mid 700 kg : $M = 8.58 \text{ kN} \times 2.320 \text{ m} = 19.91 \text{ kN.m}$ <-- governs

Far 300 kg : $M = 3.68 \text{ kN} \times 3.300 \text{ m} = 12.14 \text{ kN.m}$

Governing design moment: 19.91 kN·m at 700 kg / 2320 mm. The column, the base weld, the boom outer section and the stability check are all worst at this point.

6. Free Body and Design Forces

The factored tip load reacts through the boom into the luffing cylinder and the pivot, and from there down the column to the base and the wheels. The counterweight and crane self-weight act at the rear (free-body figure on the title page).

6.1 Cylinder and pivot forces

Taking moments about the boom pivot gives the cylinder force needed to hold the boom. The cylinder acts at a short lever arm relative to the load, so its force is large. The pivot then carries the vector sum of the cylinder force and the load reaction. The cylinder force is set by boom statics and does not depend on the material.

Boom moment balance about the pivot

Cylinder force $F_{cyl} = M_{worst} / a_{cyl} = 19.91 / 0.346 = 57.49 \text{ kN}$

Pivot reaction $F_{pivot} = \text{vector sum}(F_{cyl}, W) = 63.85 \text{ kN}$

Hydraulic pressure cross-check (luffing cyl = Type 80, bore 50 mm):

Pressure required = $57.49 \text{ kN} / 1963 \text{ mm}^2 = 29.3 \text{ MPa}$ (293 bar)

(a 30 mm bore would need 813 bar -> confirms 50 mm bore is the luffer)

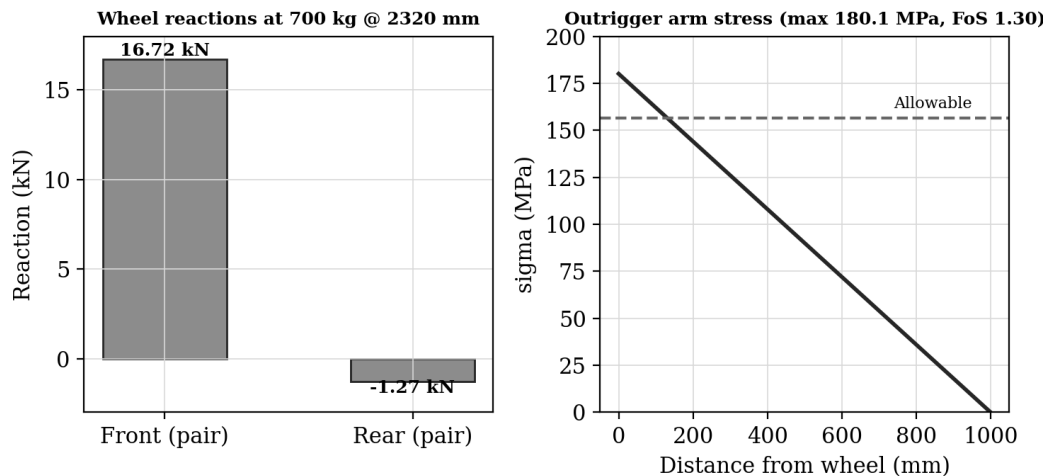
6.2 Wheel reactions

Resolving the whole crane about the wheel lines gives the support reactions. Almost all of the load goes to the front wheel pair, with the rear pair barely loaded at the governing case, because the overturning moment has nearly lifted the rear off the ground. The near-zero rear reaction confirms the crane is close to tipping.

Vertical and moment balance on the base (worst case)

$R_{front} \text{ (pair)} = 16.72 \text{ kN} \rightarrow 8.36 \text{ kN per wheel}$

$R_{rear} \text{ (pair)} = -1.27 \text{ kN} \rightarrow -0.63 \text{ kN per wheel}$



Wheel reactions (left) and outrigger-arm stress profile (right) at the governing case, Q235. The outrigger arm now reaches 180.1 MPa for a factor of 1.30, below the 156.7 MPa allowable line.

PART II — Wheels-only configuration (outriggers stowed)

7. Boom Bending

The telescopic boom steps down in three sections from root to tip. Each section is checked at its own root, where its local moment is highest. The outer section at the column end carries the full design moment. On Q235 the outer section now falls below target.

Boom bending (Q235)

Boom outer (RSH 100 x 140 x 6) at root

$M = 19.91 \text{ kN.m}$ $S_x = 107.0 \text{ cm}^3$

$\sigma = 186.2 \text{ MPa}$ $\text{FoS} = 235 / 186.2 = 1.26$ (Fail)

Boom middle (RSH 80 x 120 x 6): $\sigma = 108.1 \text{ MPa}$ $\text{FoS} = 2.17$ (Pass)

Boom inner (RSH 60 x 100 x 6): $\sigma = 89.0 \text{ MPa}$ $\text{FoS} = 2.64$ (Pass)

The boom outer section is the second structural item, after the column and weld, to fall below target on Q235; on S275 it was a marginal 1.48.

8. Column under Combined Load

The column carries axial load from the boom system plus the full bending moment transferred at its top. The two stresses add directly. This is the governing structural check, identical in both stances.

Column combined check (Q235)

Column (RSH 80 x 120 x 6), combined axial + bending

$N = 10.64 \text{ kN}$ $A = 22.6 \text{ cm}^2$ $M = 19.91 \text{ kN.m}$ $S_x = 73.0 \text{ cm}^3$

$\sigma_{\text{axial}} = 4.7 \text{ MPa}$ $\sigma_{\text{bend}} = 272.7 \text{ MPa}$

$\sigma_{\text{total}} = 277.4 \text{ MPa}$

$\text{FoS} = 235 / 277.4 = 0.85$ -> Fail (past yield)

Column buckling (Euler, $K=2$): $\lambda = 80.6$, $P_{cr} = 720 \text{ kN}$, $\text{FoS} = 67.6$ (Pass)

The column is a bending problem, not a buckling problem. On Q235 the bending stress overshoots yield, so the factor falls to 0.85. To recover a 1.5 margin the section modulus would need to rise by about three-quarters.

9. Base Frame and Outrigger Arm

The outrigger arm is checked as a cantilever picking up the front wheel reaction at its tip; the moment at the root gives the stress. On Q235 it falls below target.

Outrigger arm (Q235)

Outrigger arm (RSH 60 x 100 x 6)

$M_{\text{arm}} = 8.19 \text{ kN.m}$ $S_x = 45.5 \text{ cm}^3$

$\sigma = 180.1 \text{ MPa}$ $\text{FoS} = 235 / 180.1 = 1.30$ -> Fail

On S275 this member passed at 1.53; on Q235 the weaker steel drops it to 1.30, below target.

10. Pins and Connections

The pins are checked in shear (double or single as built) and in bearing on the lug. The cylinder force and pivot reaction drive the pin loads. On Q235 the bearing allowable falls with the steel, so the pin factors are lower than on S275.

Pin	FoS	Target	Verdict
Boom pivot pin, Ø30, double shear	2.08	3.00	Fail*
Cylinder rod-end pin, Ø20 as modelled	1.03	3.00	Fail
Cross-check at Base Detail-4 pin, Ø40	4.11	3.00	Pass
Hook pin, Ø20, single shear	3.44	3.00	Pass

* On the conservative series-factor basis coded, the pivot pin reads 2.08. The basis-independent finding is that a 20 mm rod-end pin is undersized for the 57 kN it would carry; the 40 mm Detail-4 pin clears the check on either basis. The summary carries the conservative 20 mm figure. The as-built diameter should be confirmed.

11. Welds

Fillet welds are checked on the throat area at 6 mm leg. The column-base weld carries the full moment as a stress couple across the weld group and is the governing weld; it governs alongside the column.

Weld checks (Q235, f_u 370 MPa)

Column-to-base weld (6 mm fillet)	$\sigma = 234.9$ MPa	FoS = 0.73	-> Fail
Boom-bracket weld (6 mm fillet)	$\tau = 35.5$ MPa	FoS = 2.92	-> Pass
Outtrigger weld (6 mm fillet)	$\sigma = 155.0$ MPa	FoS = 1.11	-> Fail

The column-base weld at 0.73 is the lowest factor of any load-proportional item and sets the re-rate in the companion report.

12. Slewing-Flange Bolts

The five M16 grade 8.8 bolts sit on a 170 mm pitch circle. The overturning moment is shared as a tension couple across the pattern using the polar-moment method. The bolts are Grade 8.8 fasteners, so their capacity does not change with the base-steel grade.

Slewing-flange bolts

5 x M16 grade 8.8 on 170 mm PCD, polar-moment method
 Bolt capacity : tension 113.0 kN, shear 75.4 kN
 Peak demand : tension 93.7 kN, shear 2.1 kN per bolt
 Interaction utilisation = 0.35 -> FoS (combined) = 1.69 -> Pass

13. Stability against Overturning (wheels)

Stability is a moment balance about the front wheel line, 980 mm ahead of the column. The restoring moment comes from the crane self-weight and counterweight behind that line; the overturning moment

from the load ahead of it. Stability does not depend on the material, so these figures are unchanged from the S275 study.

Overturning balance about the front wheel line

Restoring $M_R = m_{\text{crane}} \times g \times (L_{\text{fwd}} + e_{\text{cg}}) = 700 \times 9.81 \times 1.280 = 8.79 \text{ kN}\cdot\text{m}$

Overturning $M_{OT} = m_{\text{load}} \times g \times (\text{reach} - L_{\text{fwd}})$ [static load]

Load case	M_OT (kN·m)	FoS = M_R / M_OT	Verdict
1000 kg @ 1400 mm	4.12	2.13	Pass
700 kg @ 2320 mm	9.20	0.96	Fail
300 kg @ 3300 mm	6.83	1.29	Fail

Worst stability factor 0.96 at 700 kg / 2320 mm against a target of 1.50, a fail in the wheels-only state.

14. Frame Model Results

The plane-frame solver was run for all three chart points. The model confirms the column as the most-stressed member, in agreement with the hand calculation. The FoS values below are quoted against yield directly (235 / peak stress) on Q235.

Chart point	Tip deflection (mm)	Peak stress (MPa)	FoS
1000 kg @ 1400 mm	51.5	235.1	1.00
700 kg @ 2320 mm	102.7	272.7	0.86
300 kg @ 3300 mm	101.1	166.2	1.41

On Q235 the governing-case peak stress of 272.7 MPa sits above the 235 MPa yield, so the frame FoS is 0.86, matching the hand-calc 0.85. The two methods agree, as Section 16 shows.

15. Operating Envelope

The model also sweeps the boom through its luffing range and tracks the boom-root stress and factor of safety against the chart capacity at each angle. On Q235 the boom-root stress reaches the 156.7 MPa allowable across more of the luffing range than on S275, so the safe capacity drops sooner as reach grows.

16. Cross-Validation of the Two Methods

The hand calculation and the frame model are compared on the three quantities they both produce. The column-base moment matches exactly, the peak stress agrees to within two percent, and the tip deflection differs because the two measure different things.

Quantity	Track 1 (hand)	Track 2 (frame)	Agreement
Column-base moment (kN·m)	19.91	19.91	Exact
Peak member stress (MPa)	277.4	272.7	Within 2%
Boom tip deflection (mm)	22.7	102.7	Different basis

The deflection gap is expected and not an error. The hand figure is boom bending alone; the frame figure adds the column bending and the base flexibility. For a stress verdict the moment and stress agreement is what matters, and that is tight.

17. Elastic and Plastic Capacity of the Column

The column reaches first-fibre yield at a working-stress factor of safety of 0.85 on Q235, which means the applied moment at 1.25 times the rated load is above the first-yield moment of the section. First yield, however, is not collapse. The collapse limit for a compact steel section is the plastic moment, which is larger than the first-yield moment by the shape factor, 1.23 for this hollow section. That difference is reserve the elastic working-stress check does not credit. Physical load testing is carried out separately with the approval officer and is not modelled here.

Column elastic-to-plastic ladder (Q235)

Column RSH 80 x 120 x 6, elastic-to-plastic capacity (nominal Q235)

$S_x = 73.0 \text{ cm}^3$ $Z_x = 89.7 \text{ cm}^3$ shape factor $Z_x/S_x = 1.23$

applied moment @ 1.25 x SWL = 19.91 kN.m (elastic FoS 0.85)

first-yield moment $M_y = S_x \times f_y = 17.16 \text{ kN.m}$

plastic moment M_p (nom 235) = $Z_x \times f_y = 21.08 \text{ kN.m}$ (reserve over M_y +23%)

plastic moment M_p (mill 265) = 23.77 kN.m (reserve over applied +19%)

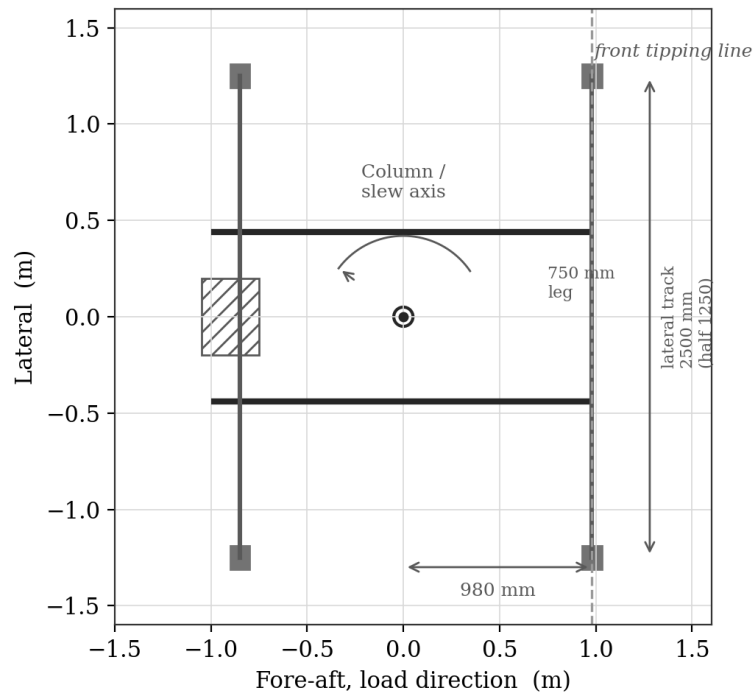
On nominal Q235 the plastic moment is 21.08 kN·m against the applied 19.91 kN·m, so the section retains a small reserve beyond first yield. On realistic mill steel the margin is wider. The factor of safety of 0.85 is a first-yield working-stress result, not a prediction of collapse; what the structure does not have, as built, is the conventional 1.5 working-stress margin at mid reach for repeated service.

PART III — Deployed-outrigger configuration (# arrangement)**18. The Deployed Configuration**

The crane base is a pair of fore-aft rails with a wheel at each corner. Deploying the legs adds two transverse beams, each reaching to both sides. The four feet, with the two rails, make the # seen in plan. The leg dimensions are read from Base DETAIL-11: each leg arm is 750 mm long, with a 230 mm drop to an 80 mm foot pad and a Ø32 hinge pin. The front foot stays level with the front frame line, 980 mm ahead of the column.

The deployed lateral track is taken from the updated general-arrangement plan view at 2500 mm foot-to-foot, with a half-track of 1250 mm. This supersedes the 2376 mm derived in earlier revisions from the bare leg-arm projection. The inner frame width is dimensioned at 1100 mm. The 2500 mm track governs sideways stability.

Quantity	Value	Source
Leg arm (lateral reach)	750 mm	DETAIL-11
Foot drop / pad	230 mm / 80 mm	DETAIL-11
Leg hinge pin	Ø32	DETAIL-11
Inner frame width	1100 mm	GA plan view
Deployed lateral track	2500 mm (foot-to-foot)	GA plan view
Front foot ahead of column	980 mm (front frame line)	Base drawing + warning plate
Leg member section	RHS 60 x 100 x 6	drawing
Material	Q235 (fy 235 MPa)	client-confirmed

Deployed footprint, plan view: two rails plus four transverse legs (# arrangement

Deployed footprint in plan. The legs give a 2500 mm foot-to-foot track, while the front foot stays at the 980 mm front line.

19. Forward Stability (deployed)

Forward tipping is a moment balance about the front foot line, 980 mm ahead of the column. With the legs at 90 degrees, that line is in the same place as the front wheel, so the balance is identical to the wheels-only case. Transverse legs do not move the front line, so they cannot improve forward stability, and the material does not enter a stability balance.

Load case	M_OT (kN·m)	FoS = M_R / M_OT	Verdict
1000 kg @ 1400 mm	4.12	2.13	Pass
700 kg @ 2320 mm	9.20	0.96	Fail
300 kg @ 3300 mm	6.83	1.29	Fail

The mid-reach point governs at 0.96, below target, matching the wheels-only result as it must.

20. Sideways Stability During Slew

Sideways tipping is the case the legs are there for. With the boom slewed across the rails, the crane can tip over a side foot line set at the half-track, 1250 mm from the centreline on the 2500 mm track. The restoring moment is the crane weight at that half-track; the overturning moment is the lifted load at its reach less the half-track.

Overturning balance about a side foot line

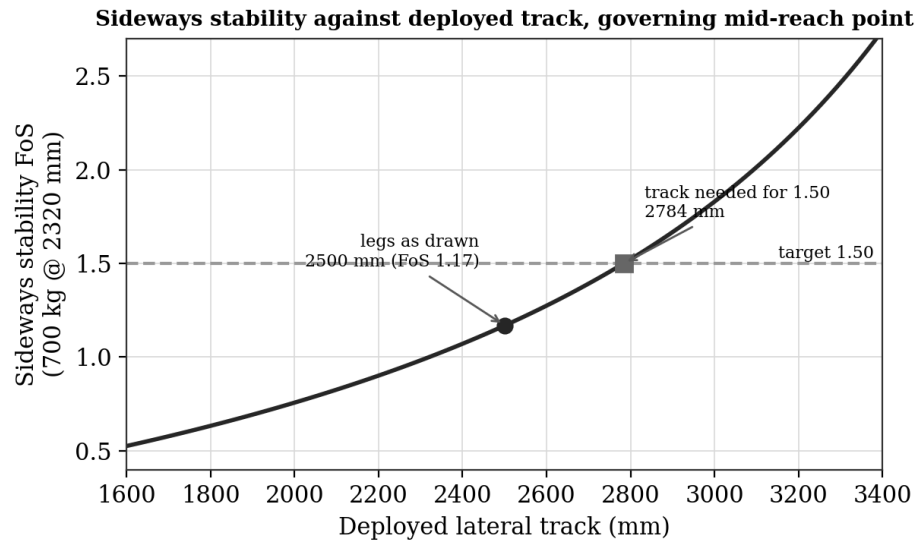
Restoring $M_R = m_{\text{crane}} \times g \times t_{\text{half}} = 700 \times 9.81 \times 1.250 = 8.58 \text{ kN.m}$

Overturning $M_{OT} = m_{\text{load}} \times g \times (\text{reach} - t_{\text{half}})$ [static load]

Half-track from the drawing: $t_{\text{half}} = 2500 / 2 = 1250 \text{ mm}$ (track 2500 mm)

Load case	M_OT (kN·m)	FoS = M_R / M_OT	Verdict
1000 kg @ 1400 mm	1.47	5.83	Pass
700 kg @ 2320 mm	7.35	1.17	Fail
300 kg @ 3300 mm	6.03	1.42	Fail

The close-in lift is steady, but the mid and far points fall below target, at 1.17 and 1.42. Solving the balance for 1.50 at the governing mid point gives a half-track of 1392 mm, a full track of 2784 mm. The 2500 mm drawn track is about 284 mm short of that; the deployed width should be confirmed and, if the feet adjust, set wider before slewing at mid or long reach. The 2500 mm track is an improvement on the 2376 mm of earlier revisions but does not by itself clear the target.



Sideways stability against the deployed track at the governing mid-reach point. The 2500 mm track sits below the 1.50 target; about 2784 mm is needed.

21. The Deployed Leg Member

The deployed leg is checked as a cantilever picking up the vertical reaction at its foot. At 1000 kg and 1400 mm the front reaction is 17.43 kN, shared by the two front feet.

Deployed leg, RHS 60 x 100 x 6, cantilever (Q235)

front reaction (1000 kg @ 1400 mm) = 17.43 kN -> 8.72 kN per leg
 lever (leg arm) = 0.750 m
 $M_{leg} = 8.72 \times 0.750 = 6.54 \text{ kN.m}$ $S_x = 45.5 \text{ cm}^3$
 $\sigma = 143.7 \text{ MPa}$ $FoS = 235 / 143.7 = 1.64$ -> Pass

The leg passes at 1.64 on Q235, down from 1.91 on S275 but still above target. It is less stressed than the wheels-only outrigger arm, which carries the front reaction over the longer 980 mm fore-aft cantilever and now fails at 1.30 on the weaker steel.

22. Items Carried Over from the Wheels-Only Check

The legs attach to the base frame, below the slewing bearing. The axial load and bending moment that pass up into the column, the boom and its pins are fixed by the boom geometry and the lifted load, and do not change when the base stands on legs instead of wheels. Every stress-based factor below is on Q235.

Check	FoS	Target	Verdict
Boom outer (100x140x6)	1.26	1.50	Fail
Boom middle (80x120x6)	2.17	1.50	Pass
Boom inner (60x100x6)	2.64	1.50	Pass
Column, combined N+M	0.85	1.50	Fail
Column buckling	67.6	1.50	Pass
Base outrigger arm (fore-aft cantilever)	1.30	1.50	Fail
Pivot pin (Ø30)	2.08	3.00	Fail
Cylinder rod-end pin (Ø20 as modelled)	1.03	3.00	Fail
Hook pin (Ø20)	3.44	3.00	Pass
Slewing bolts (5 x M16, 8.8)	1.69	1.50	Pass
Weld, column base	0.73	1.50	Fail
Weld, boom bracket	2.92	1.50	Pass

The column at 0.85 and its base weld at 0.73 remain the governing structural items in both configurations.

23. Stowed versus Deployed

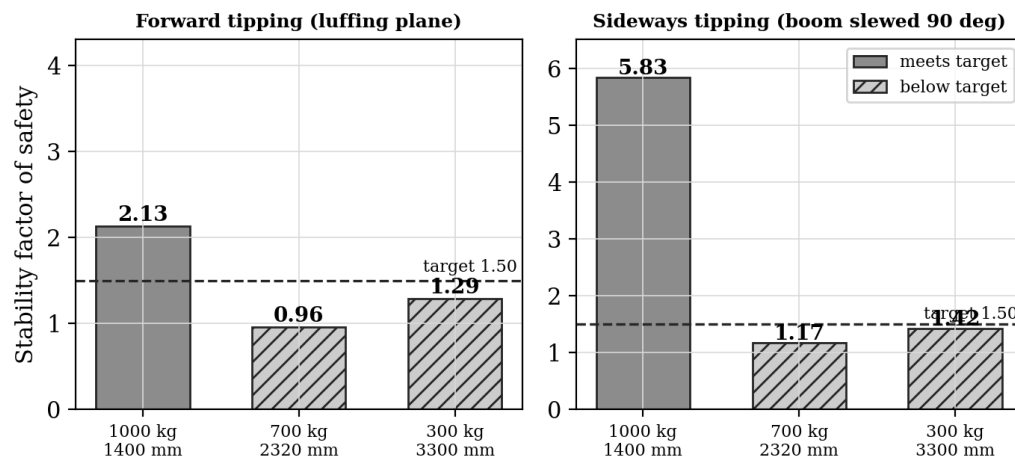
The legs change the bottom of the load path and the stability picture; they leave the top of the structure exactly as it was, and the Q235 material acts on both states alike.

Item	Wheels stowed	Legs deployed (# arrangement)
Front tipping line ahead of column	980 mm	980 mm (no forward reach)
Lateral foot track	876 mm	2500 mm

Item	Wheels stowed	Legs deployed (# arrangement)
Forward tipping, worst point	FoS 0.96, Fail	FoS 0.96, Fail (unchanged)
Sideways tipping during slew	narrow track, tips	FoS 1.17, Fail (track short)
Front ground-support member	outrigger arm 180 MPa, FoS 1.30, Fail	leg arm 144 MPa, FoS 1.64, Pass
Column, combined N+M	277 MPa, FoS 0.85, Fail	unchanged
Column-to-base weld	235 MPa, FoS 0.73, Fail	unchanged
Boom outer section	186 MPa, FoS 1.26, Fail	unchanged
Pins and slewing bolts	Q235 base, Gr 8.8 bolts	unchanged

Read across, the pattern is consistent: the legs widen the track and ease the leg member, so sideways tipping becomes possible to manage and the leg passes; forward tipping, the column, the weld, the boom and the outrigger arm are the same in both states, because the legs do not act on them and the material is common.

Stability with the legs deployed (# arrangement, 90 deg transverse)



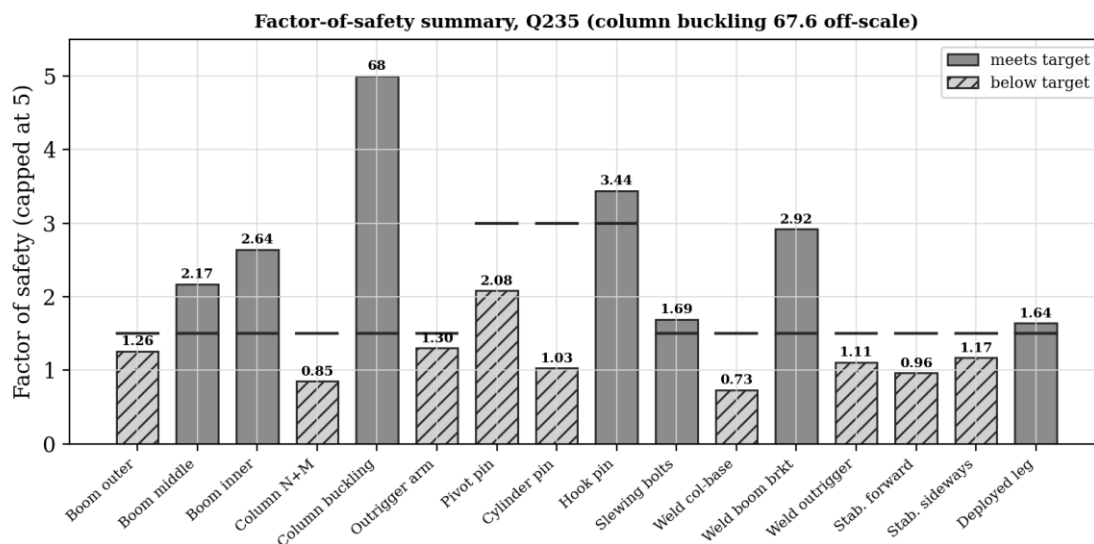
Stability in the deployed state on the 2500 mm track. Forward tipping is unchanged from the wheels-only case; sideways tipping clears the target only close in.

PART IV — Combined outcome

24. Summary of Results

All sixteen checks are collected below with their factors of safety and verdicts against target, on Q235. Ten of the sixteen do not meet target on a working-stress basis. The column and its base weld govern; the boom outer and the outrigger arm now fail on the weaker steel. Stability is shown for both stances.

Check	FoS	Target	Verdict
Boom outer (100x140x6)	1.26	1.50	Fail
Boom middle (80x120x6)	2.17	1.50	Pass
Boom inner (60x100x6)	2.64	1.50	Pass
Column, combined N+M	0.85	1.50	Fail
Column buckling	67.62	1.50	Pass
Outrigger arm	1.30	1.50	Fail
Pivot pin	2.08	3.00	Fail
Cylinder pin (Ø20 as modelled)	1.03	3.00	Fail
Hook pin	3.44	3.00	Pass
Slewing bolts	1.69	1.50	Pass
Weld, column base	0.73	1.50	Fail
Weld, boom bracket	2.92	1.50	Pass
Weld, outrigger	1.11	1.50	Fail
Stability, forward (both stances)	0.96	1.50	Fail
Stability, sideways slew (2500 track)	1.17	1.50	Fail
Deployed leg member (RHS 60x100x6)	1.64	1.50	Pass



Factor of safety for every check against the 1.5 / 3.0 targets, Q235. Hatched bars are below target; the black tick marks the required factor. Column buckling (67.6) is off-scale.

25. Conclusions and Recommendations

25.1 Conclusion on the 1 tonne SWL

In neither stance does the crane meet the agreed factors of safety across its rated chart, and on Q235 the margin is thinner than the earlier S275 study showed. On its wheels, the column reaches yield (0.85) and its base weld is well past target (0.73) at the governing mid-reach point, and the boom outer and outrigger arm now fall below target. Deploying the legs makes slewing possible and eases the front-support member, but it does not touch the column, the weld or forward tipping. The crane cannot be certified to the full 1 tonne chart on a working-stress basis in either configuration.

25.2 The governing items

- Column, FoS 0.85 (both stances). Past yield in bending; the clear primary item, though it retains a small plastic reserve as Section 17 shows.
- Column-base weld, FoS 0.73 (both stances). The lowest factor of any load-proportional item; addressed alongside the column and sets the re-rate.
- Boom outer and base outrigger arm, FoS 1.26 and 1.30. Both fail on Q235 where they were marginal or passing on S275. The re-rate that fixes the column also brings these back above target.
- Sideways track, 2500 mm against 2784 mm needed. The one item the deployed configuration itself must address; confirm the deployed width before slewing at mid or long reach.
- Cylinder rod-end pin, FoS 1.03 (20 mm). Undersized if 20 mm; the 40 mm Detail-4 pin passes at 4.11. Confirm the as-built size.

25.3 Recommendations

Route	Outcome
Re-rate the chart (recommended)	Reduce the rated capacities so every working-stress check clears target. The separate Re-Rating Report (Rev R2) sets a maximum SWL of 560 kg, with 340 kg at mid reach and 240 kg at far reach on Q235. Cheapest route, no fabrication, applies in both stances.
Strengthen the column and base (keeps full rating)	A larger column section, with the base weld grown to suit and the cylinder pin confirmed or upsized. The only route that preserves the full 1 tonne chart; on Q235 the upgrade is larger than the S275 case would have needed.
Confirm mill yield, cylinder geometry and deployed track	Parallel actions. Confirming actual Q235 mill yield and the cylinder anchor would recover some column margin; confirming the deployed track clears sideways stability.

Recommendation. Adopt the re-rate as the immediate action so the crane can be used safely at a reduced, verified rating, and hold the column upgrade as the path to retaining the full 1 tonne chart. Deploy the legs before any slew. The crane should not be worked to the current 1 tonne chart until the column, base weld, boom outer, outrigger arm and cylinder pin are addressed.

26. Assumptions and Items to Confirm

- Material, Q235. Client-confirmed, fy 235 MPa nominal; no thickness de-rating (6 mm walls). Mill certificates would confirm the actual yield, which commonly runs a little above nominal.

- Crane self-weight, 700 kg, and CG offset, 0.30 m behind the column. Client-supplied and estimated; a swing of ± 100 kg moves both stability results.
- Deployed lateral track, 2500 mm, and leg angle, 90 degrees. From the updated drawing and the deployment schematic. The as-built width and angle must be measured; a forward splay would improve forward stability.
- Front overhang to the tipping line, 980 mm. From the base top-view chain and the warning-plate 420 mm dimension. The 2360 mm wheelbase on the new drawing is unchanged.
- Cylinder lever arm (0.346 m) and rod-end pin diameter. The geometry reproduces the issued cylinder force but is not dimensioned; the pin diameter is to be confirmed.
- Weld throats, 6 mm, the drawing minimum. If site welds are smaller the governing weld worsens.

27. Standards Referenced

- EN 13001-3-1, crane structural design, load effects and proof of competence
- EN 1993-1-1, design of steel structures, general rules
- EN 1993-1-8, design of joints
- GB/T 700, carbon structural steels (Q235)
- Cross-referenced with AS 1418.1 and SS 559 for the Singapore context

Appendix A: Companion files

- Load Re-Rating Report (Rev R2), which converts these findings into a reduced SWL chart on Q235.
- MATLAB R2026a parametric model (project 40467361), on Q235, the single file driving every number in this report.
- Console logs of the wheels-only (C1_v3) and deployed (C2_v3) runs, including the factor-of-safety tables and the column plastic-capacity check.